

Classification: Naïve Bayes

Natural Language Processing

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Probability – Recap

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The number of students enrolled in different programs at a University is given below:

	Male	Female	Total
Engineering	64	16	80
Medicine	22	71	93
Total	86	87	173

- What are the chances that a randomly selected student of this university is:

a) of Engineering?

$$80/173 = \underline{\hspace{2cm}}$$

b) a female?

$$87/173 = \underline{\hspace{2cm}}$$

c) a male and in Medicine?

$$= \frac{22}{173} = \underline{\hspace{2cm}}$$

	B		~B	
	C	~C	C	~C
A	12	5	9	2
~A	<u>4</u>	8	<u>20</u>	4

- $P(A \mid B, C) = \underline{12/16}$
- $P(A, B \mid \underline{\sim C}) = 5 / 19$
- $P(\underline{B} \mid \sim A, C) = 4 / \underline{24}$

Dependent/Independent Events

- If the occurrence of Event A changes the probability of Event B, then Events A and B are **dependent**. On the other hand, if the occurrence of Event A does not change the probability of Event B, then Events A and B are **independent**.

Conditional Independence

- Two events A and B are independent if knowing that A has happened does not say anything about B happening.

$$P(A \text{ B}) = P(A) P(B)$$

$$P(A \mid B) = P(A)$$

- Two events A and B are conditionally independent given a third event C precisely if the occurrence or non-occurrence of A and B are independent events in their conditional probability distribution given C .

$$P(A \text{ B} \mid C) = P(A \mid C) P(B \mid C)$$

$$P(A \mid B \text{ C}) = P(A \mid C)$$

Naïve Bayes

Prior and Posterior probability

- The prior probability is the initial probability of an event before it is contextualized under a certain condition, or the marginal probability.
 - $P(A)$
- The posterior probability is the probability of an event after observing a piece of data.
 - $P(A|B)$

Example (medical testing)

$$p(\text{diabetes} \mid + \text{ test}) = \frac{p(+ \text{ test} \mid \text{diabetes})p(\text{diabetes})}{P(+ \text{ test} \mid \text{diabetes})p(\text{diabetes}) + p(+ \text{ test} \mid \text{no diabetes}) p(\text{no diabetes})}$$

Bayes Rule

- Bayes' rule gives us a way to break down any conditional probability $P(x|y)$ into three other, where $P(x|y)$ is posterior probability
- $P(A|B) = P(B|A)P(A)/P(B)$

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

Example I

- According to American Lung Association, 7% of the population has lung cancer. Of these people having lung disease, 90% are smokers; and of those not having lung disease, 25.3% are smokers.

- Determine the probability that a randomly selected smoker has lung cancer.
 $L \rightarrow \text{Lung Disease}$ $S \rightarrow \text{Smoker}$
 $P(L) = 0.07$ $P(S|L) = 0.9$ $P(S|\bar{L}) = 0.253$
 $P(\bar{L}) = 1 - 0.07 = 0.93$
 $P(L|S) = ?$ $P(L|S) = \frac{P(S|L)P(L)}{P(S|L)P(L) + P(S|\bar{L})P(\bar{L})}$
 $= \frac{0.9 \cdot 0.07}{0.9 \cdot 0.07 + 0.253 \cdot 0.93}$

Example I Solution

- Let L = Lung Cancer, S = Smoker
- Given that
 - $P(L) = 0.07$
 - $P(S | L) = 0.90$ $P(\sim S | L) = 0.10$
 - $P(S | \sim L) = 0.253$ $P(\sim S | \sim L) = 0.747$
- Find probability, $P(L | S)$

$$P(L | S) = \frac{P(S \cap L)}{P(S)} = \frac{P(S | L)P(L)}{P(S | L)P(L) + P(S | \sim L)P(\sim L)}$$
$$P(L | S) = \frac{0.9 \times 0.07}{0.9 \times 0.07 + 0.253 \times 0.93} = \underline{\hspace{2cm}}$$

Naïve Bayes Classifier

- Naive Bayes is a probabilistic classifier, meaning that for a document d , out of all classes $c \in C$ the classifier returns the class $\hat{c}$ which has the maximum posterior probability given the document.

$$\hat{c} = \operatorname{argmax}_{c \in C} P(c|d)$$

$$\hat{c} = \operatorname{argmax}_{c \in C} P(c|d) = \operatorname{argmax}_{c \in C} \frac{P(d|c)P(c)}{P(d)}$$

$$\hat{c} = \operatorname{argmax}_{c \in C} P(c|d) = \operatorname{argmax}_{c \in C} \frac{P(d|c)P(c)}{P(d)}$$

$$\hat{c} = \operatorname{argmax}_{c \in C} P(c|d) = \operatorname{argmax}_{c \in C} P(d|c)P(c)$$

$$\hat{c} = \operatorname{argmax}_{c \in C} \overbrace{P(d|c)}^{\text{likelihood}} \overbrace{P(c)}^{\text{prior}} \quad (4.5)$$

Without loss of generalization, we can represent a document d as a set of features $f_1, f_2, \dots, f_n$:

$$\hat{c} = \operatorname{argmax}_{c \in C} \overbrace{P(f_1, f_2, \dots, f_n|c)}^{\text{likelihood}} \cdot \overbrace{P(c)}^{\text{prior}} \quad (4.6)$$

$$P(f_1, f_2, \dots, f_n | c) = P(f_1 | c) \cdot P(f_2 | c) \cdot \dots \cdot P(f_n | c) \quad (4.7)$$

The final equation for the class chosen by a naive **Bayes** classifier is thus:

$$c_{NB} = \operatorname{argmax}_{c \in C} P(c) \prod_{f \in F} P(f | c) \quad (4.8)$$

Naïve Bayes for text classification

positions $\leftarrow$ all word positions in test document

$$c_{NB} = \operatorname{argmax}_{c \in C} P(c) \prod_{i \in \text{positions}} P(w_i | c) \quad (4.9)$$

$$c_{NB} = \operatorname{argmax}_{c \in C} \log P(c) + \sum_{i \in \text{positions}} \log P(w_i | c)$$

	Cat	Documents
Training	-	just plain boring
	-	entirely predictable and lacks energy
	-	no surprises and very few laughs
	+	very powerful
	+	the most fun film of the summer
Test	?	predictable with no fun

$$P(-) = \frac{3}{5} \quad P(+) = \frac{2}{5}$$

Bag of words

- [just, plain, boring, entirely, predictable, and, lacks, energy, no, surprises, very, few, laughs, powerful, the, most, fun, film, of, summer]
- $|V|=20$

- Test: Predictable with no fun
- Test: Predictable no fun
- [just, plain, boring, entirely, predictable, and, lacks, energy, no, surprises, very, few, laughs, powerful, the, most, fun, film, of, summer]
- [0,0,0,0,1,0,0,0,1,0,0,0,0,0,0,0,1,0,0,0]

$$P(\text{“predictable”}|-) = \frac{1+1}{14+20} \quad P(\text{“predictable”}|+) = \frac{0+1}{9+20}$$

$$P(\text{“no”}|-) = \frac{1+1}{14+20} \quad P(\text{“no”}|+) = \frac{0+1}{9+20}$$

$$P(\text{“fun”}|-) = \frac{0+1}{14+20} \quad P(\text{“fun”}|+) = \frac{1+1}{9+20}$$

- S belongs to a –ve (negative)Class.

For the test sentence $S = \text{“predictable with no fun”}$, after removing the word ‘with’, the chosen class, via Eq. 4.9, is therefore computed as follows:

$$P(-)P(S|-) = \frac{3}{5} \times \frac{2 \times 2 \times 1}{34^3} = 6.1 \times 10^{-5}$$

$$P(+)P(S|+) = \frac{2}{5} \times \frac{1 \times 1 \times 2}{29^3} = 3.2 \times 10^{-5}$$

Multinomial naive Bayes or binary NB

	NB Counts		Binary Counts		
	+	−	+	−	
Four original documents:					
− it was pathetic the worst part was the boxing scenes	and	2	0	1	0
	boxing	0	1	0	1
	film	1	0	1	0
− no plot twists or great scenes	great	3	1	2	1
+ and satire and great plot twists	it	0	1	0	1
+ great scenes great film	no	0	1	0	1
	or	0	1	0	1
	part	0	1	0	1
	pathetic	0	1	0	1
	plot	1	1	1	1
	satire	1	0	1	0
	scenes	1	2	1	2
	the	0	2	0	1
	twists	1	1	1	1
	was	0	2	0	1
	worst	0	1	0	1
After per-document binarization:					
− it was pathetic the worst part boxing scenes					
− no plot twists or great scenes					
+ and satire great plot twists					
+ great scenes film					

Figure 4.3 An example of binarization for the binary naive Bayes algorithm.

Evaluation

Accuracy

- Ratio of the number of correct predictions to the total number of input samples.
- $\text{Accuracy} = \frac{\text{Correct prediction}}{\text{Total prediction}}$

		<i>gold standard labels</i>		
		gold positive	gold negative	
<i>system output labels</i>	system positive	true positive	false positive	precision = $\frac{tp}{tp+fp}$
	system negative	false negative	true negative	
		recall = $\frac{tp}{tp+fn}$		accuracy = $\frac{tp+tn}{tp+fp+tn+fn}$

Figure 4.4 Contingency table

Precision

- Precision measures the percentage of the items that the system detected (i.e., the system labeled as positive) that are in fact positive (i.e., are positive according to the human gold labels). Precision is defined as:

$$\mathbf{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

Recall

- Recall measures the percentage of items actually present in the input that were correctly identified by the system. Recall is defined as:

$$\mathbf{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

F1-Score

- a single metric that incorporates aspects of both F-measure precision and recall.

$$F_1 = \frac{2PR}{P+R}$$

Example

- Suppose we have a binary classification task, such as detecting spam emails (positive class = spam, negative class = not spam). Let's assume the model makes predictions for 10 emails, and all predictions are correct.
 - **True Positives (TP):** 5 emails are correctly predicted as spam.
 - **True Negatives (TN):** 5 emails are correctly predicted as not spam.
 - **False Positives (FP):** 0 emails are incorrectly predicted as spam.
 - **False Negatives (FN):** 0 emails are incorrectly predicted as not spam.

Calculate

- Precision
- Recall
- F1-score

Now, let's calculate the metrics:

1. **Precision:** Since all predicted positives are correct (no false positives),

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{5}{5 + 0} = 1$$

2. **Recall:** Since all actual positives are correctly predicted (no false negatives),

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{5}{5 + 0} = 1$$

3. **F1-score:** Using the formula for F1-score,

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \times \frac{1 \times 1}{1 + 1} = 1$$

Since both precision and recall are perfect, the F1-score is 1, indicating ideal performance by the model.



F-measure

F-measure There are many ways to define a single metric that incorporates aspects of both precision and recall. The simplest of these combinations is the **F-measure** (van Rijsbergen, 1975), defined as:

$$F_{\beta} = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

The β parameter differentially weights the importance of recall and precision, based perhaps on the needs of an application. Values of $\beta > 1$ favor recall, while values of $\beta < 1$ favor precision. When $\beta = 1$, precision and recall are equally balanced; this is the most frequently used metric, and is called $F_{\beta=1}$ or just F_1 :

1. Conditional Probability:

Conditional probability is the probability of an event occurring given that another event has already occurred. It is represented as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Here:

- $P(A|B)$ is the probability of event A occurring, given that event B has occurred.
- $P(A \cap B)$ is the joint probability that both events A and B occur.
- $P(B)$ is the probability of event B occurring.

2. Posterior Probability:

Posterior probability is a specific kind of conditional probability that arises in **Bayesian statistics**. It refers to the probability of a hypothesis or event after incorporating new evidence or information. The posterior probability is updated based on prior knowledge (prior probability) and the likelihood of the observed data.

The formula for posterior probability using **Bayes' Theorem** is:

$$P(H|D) = \frac{P(D|H) \times P(H)}{P(D)}$$

Where:

- $P(H|D)$ is the **posterior probability**: the probability of the hypothesis H given the data D (new evidence).
- $P(D|H)$ is the **likelihood**: the probability of observing the data D given that the hypothesis H is true.
- $P(H)$ is the **prior probability**: the initial belief about the probability of hypothesis H before observing the data.
- $P(D)$ is the **evidence**: the total probability of observing the data, considering all possible hypotheses.